Biprateep Dey

Curriculum Vitae

Department of Statistical Sciences,
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Employment

2024-present **Eric and Wendy Schmidt AI in Science Postdoctoral Fellow**, *Department of Statistical Sciences*, University of Toronto & Vector Institute for Artificial Intelligence, Toronto, Ontario, Canada

2024-present **CITA Postdoctoral Fellow**, *Canadian Institute for Theoretical Astrophysics (CITA)*, University of Toronto, Toronto, Ontario, Canada

2024-present **Dunlap Fellow**, *Dunlap Institute for Astronomy and Astrophysics*, University of Toronto, Toronto, Ontario, Canada

Education

2024 **Ph.D. in Physics**, *Department of Physics and Astronomy, University of Pittsburgh*, Pittsburgh, Pennsylvania, USA
Thesis Title: Cosmic cartography: Photometric Redshifts for the Next-Generation of Sky Surveys
Advisors: Prof. Jeff Newman and Prof. Brett Andrews

2020 **M.S. in Physics**, *University of Pittsburgh*, Pittsburgh, Pennsylvania, USA

2018 **Integrated B.Sc.-M.Sc. in Physics**, *National Institute of Science Education and Research (NISER)*, Bhubaneswar, Odisha, India
Thesis Title: Constructing predictors for HI mass in galaxies
Advisor: Prof. Nishikanta Khandai

Research Interests

Dark Matter, Galaxy Formation and Evolution, Statistical Machine Learning and Astronomical Sky Surveys

Publications: **12 lead/significant contributing author, and 127 contributing author** (list attached)

Presentations: **27 invited, 21 contributed** (list attached)

Awards and Honors

2024 **Outstanding Dissertation Award**, American Physical Society Topical Group on Data Science (APS-GDS)

2023 **"Builder" of the Dark Energy Spectroscopic Instrument (DESI) Collaboration**, Builder status is awarded to DESI members in recognition of a long engagement and a significant contribution to the collaboration infrastructure and service work

2023 **American Physical Society Topical Group on Data Science IMPACT Award for Excellence in Graduate Research**

2022 **LSST Corporation Enabling Science Fellowship**, Funding to attend the 2022 Rubin Observatory's Project and Community Workshop

2022 - 23 **Andrew Mellon Predoctoral Fellowships**, Funding for year long research at the Univ. of Pittsburgh

2022 **Zaccheus Daniel Predoctoral Fellowship**, Funding for summer term research at the Univ. of Pittsburgh

2021 **PITT-PACC Fellowship**, Funding for fall term research at the Univ. of Pittsburgh

2017 **MITACS Globalink Research Fellowship**, Funding for summer research at the Univ. of Alberta

2014 - 2018 **Kishore Vaigyanik Protsahan Yojana (KVPY) Fellowship**, *Dept. of Science and Tech., Govt. of India*, Scholarship for undergraduate studies and research internships

2013 - 2014 **INSPIRE Fellowship**, *Dept. of Science and Tech., Govt. of India*, Scholarship for undergraduate studies

Leadership & Service

Invited Referee

- National Aeronautics and Space Administration (NASA)
- Canadian Space Agency (CSA)
- Monthly Notices of the Royal Astronomical Society (MNRAS)
- Astronomy & Astrophysics (A&A)
- The Astronomical Journal (AJ)
- Neural Information Processing Systems (NeurIPS) Machine Learning and the Physical Sciences Workshop (2022-2026)
- Conference on Physics and AI at Stanford University (PAI 2026)
- International Conference on Machine Learning (ICML): Synergy of Scientific & Machine Learning Modeling Workshop (2023), Machine Learning for Astrophysics Workshop (2025)

2026 **Member**, Summer Undergraduate Research Program Committee

2026 **Member**, University of Toronto Astronomy Community and Culture Committee

2025-2026 **Organizer**, Black Board Seminar Series, Canadian Institute for Theoretical Astrophysics

2024 **Judge**, AAS 243 Chambliss Astronomy Achievement Student Awards

2023 **Member**, LSST-DESC Collaboration Council Nominating Committee

2023 - present **Full Member**, LSST Dark Energy Science Collaboration

2023-2026 **Co-Chair**, DESI photo-z Topical Group

2022 **Chair**, LSST-DESC Collaboration Council Nominating Committee

2022 - 2023 **Member**, DESI Committee for Early Career Scientists

2021 - 2022 **Chairperson**, DESI Committee for Early Career Scientists

2020 - 2022 **Member**, DESI Outreach Committee

2020 - 2021 **Coordinator**, Astrosnacks, Dept. of Physics and Astronomy, University of Pittsburgh
Organized student driven talk and tutorial series

2020 - 2021 **Secretary**, Executive Committee, Bengali Association of Pittsburgh

2019 - 2021 **Coordinator**, Astronomy on Tap, Pittsburgh

Awarded Super-computing Time

2026 **Perlmutter Supercomputer at NERSC**, *Photometric Redshifts for LSST and Roman*, **500 CPU Hours and 8000 GPU Hours**, PI: **A. Khederlarian**, Co-I:**B. Dey** and others

2025 **Perlmutter Supercomputer at NERSC**, *Photometric Redshifts for LSST and Beyond*, **500 CPU Hours and 8000 GPU Hours**, PI: **B. Dey**, Co-I: J. Newman, B. Andrews

2024 **Perlmutter Supercomputer at NERSC**, *Photometric Redshifts for LSST and Beyond*, **1500 CPU Hours and 8000 GPU Hours**, PI: **B. Dey**, Co-I: J. Newman, B. Andrews

2023 **Perlmutter Supercomputer at NERSC**, *Photometric Redshifts for LSST*, **500 CPU Hours and 2800 GPU Hours**, PI: **B. Dey**, Co-I: J. Newman, B. Andrews

2022 **Neocortex (Cerebras Wafer Scale Engine) at Pittsburgh Supercomputing Center** , *Making the Largest Map of Our Universe*, **500 machine hours**, PI: **B. Dey**, Co-I: J. Newman, B. Andrews, J. Rajasegaran

Awarded Telescope Time

2023 **Dark Energy Spectroscopic Instrument (DESI)**, *Testing ELG Selections for DESI-2*, **18k fiber hours**, PI: J. Newman, Co-I: **B. Dey** and others

2023 **Dark Energy Spectroscopic Instrument (DESI)**, *Four in one: A consolidated program for DESI-2 and DESI-1b science cases in the COSMOS field*, **5k fiber hours**, PI: **B. Dey**, Co-PI: A. Leauthaud, J. Newman, R. Wechsler, Y. Mao

2022 **Dark Energy Spectroscopic Instrument (DESI)**, *DESI-2 for Deep Spectroscopic Samples for LSST Photo-z's*, **6k fiber hours**, PI: **B. Dey**, Co-I: J. Newman, B. Andrews, R. Zhou, J. Myles, J. McCullough, D. Gruen, N. Weaverdyck

2022 **Hubble Space Telescope Cycle 30 SNAP Proposal**, *Post-starbursts from DESI: Timing quenching and morphological transformation at* $1 < z < 1.3$, **409 Orbits**, PI: D. Setton, Co-I: **B. Dey** and others

Awarded Funding*

2025 **Schmidt Sciences Community Initiative Fund**, *AI Foundation Models for Next-Generation Scientific Collaborations*, ∼**$100k**, Co-PI: **B. Dey** and others

2025 **Dunlap Institute & CITA Conference Funding**, *Cosmology with Lyman Break Galaxies*, ∼**$7k**, Co-PI: **B. Dey** and others

2023 **Nancy Grace Roman Space Telescope Research and Support Participation Opportunities**, *Exploiting Deep Learning to Improve Roman Photometric Redshifts*, ∼**$219k**, PI: J. Newman, Co-I: **B. Dey** and others

2023 **Nancy Grace Roman Space Telescope Research and Support Participation Opportunities**, *A Statistical Framework for Optimizing Roman Spectroscopic Training Sets*, ∼**$219k**, PI: J. Newman, Co-I: **B. Dey** and others

2022 **Hubble Space Telescope Cycle 30 SNAP**, *Post-starbursts from DESI: Timing quenching and morphological transformation at* $1 < z < 1.3$, ∼**$203k**, PI: D. Setton, Co-I: **B. Dey** and others

2021 **2022 ACCelerate Creativity + Innovation Festival**, Secured funding (∼**$12K**) from the University of Pittsburgh and the Atlantic Coast Conference (ACC) to produce a museum exhibit on *Making the largest Maps of our Universe*, **PI: B. Dey**, Co-I: J. Newman

* Monetory amounts in USD

Supervision and Mentorship

2026 **Rina Rast**, *Canadian Astronomical Society (CASCA) Postdoctoral Mentorship Program*, Mentoring for postdoctoral fellowship application preparation

Summer 2026 **Tristan Fattedad**, *Photometric Redshift Calibration for the Vera C. Rubin Observatory*, Undergraduate research project at University of Toronto

Summer 2026 **Hanbang (Leo) Wu**, *alphaUniverse: The AI Virtual Observatory*, Undergraduate research project at University of Toronto
Co-supervised with Josh Speagle and Nolan Koblischke.

2025-2026 **Alexandros Pratsos**, *Stellar Stream Identification with Machine Learning*, Undergraduate research project at University of Toronto
Co-supervised with Ting Li.

2025-2026 **Zehao Peng**, *Foundation Models for Astronomical Data*, Undergraduate research project at University of Toronto
Co-supervised with Josh Speagle and Chris Maddison

2025 **Matteo Moretti**, *Evaluating Probabilistic Photometric Redshift Estimation Algorithms*, Undergraduate research project at University of Toronto
Co-supervised with Renée Hložek.

Summer 2025 **Cissy Kuang**, *Investigating the Properties of Fast Radio Burst Host Galaxies*, Undergraduate research project at University of Toronto
Co-supervised with Aditya Vijaykumar and Maya Fishbach.

Summer 2025 **Anna Tsvetkova**, *Image Segmentation for Strongly Lensed Systems*, Undergraduate research project at University of Toronto
Co-supervised with Mike Walmsley and Josh Speagle.

2024-2025 **Isabelle Laing**, *Stellar Parameter Estimation with Deep Learning*, Undergraduate research project at University of Toronto
Co-supervised with Ting Li and Josh Speagle.

2024 **Skyler (Duo) Yang**, *Photometric Redshift Distributions for Weak Lensing Cosmology*, Undergraduate reading project at University of Toronto

2024 **Farnia Najafi-Shoushtari**, *Empirical Models of Galaxy-Halo Connection*, Undergraduate reading project at University of Toronto

2023-2024 **Emma Moran**, *Assessing Photometric Redshifts in Local Regions of Color Space*, Undergraduate research project at University of Pittsburgh
Co-supervised with Brett Andrews and Jeff Newman.

2022 - 2023 **Graduate Student Mentor**, Mentor for 5 incoming students at the Dept. of Physics and Astronomy, University of Pittsburgh

2020 - 2021 **Graduate Student Mentor**, Mentor for 3 incoming students at the Dept. of Physics and Astronomy, University of Pittsburgh

Teaching

July 2026 **A Practitioner's Guide to Uncertainty Quantification**, Presented tutorial at the STATSTRO 2026 workshop at the University of Toronto

November 2025 **Training and Fine-tuning Foundation Models**, Presented tutorial at the Foundation Models for Science Workshop at the University of Toronto

April 2024 **Calibration of Uncertainty Estimates for Astronomical Analysis**, Presented tutorial at the summer 2024 LSST-ISSC Collaboration meeting

July 2023 **DESI Spectra Visual Inspection Training**, Presented tutorial at the summer 2023 DESI Collaboration meeting

Summer 2022 **AstroPGH Bootcamp**, Presented two lectures on Astropy

Summer 2021 **AstroPGH-TAMU Bootcamp**, Presented two lectures on introductory Numpy

Summer 2020 **AstroPGH Bootcamp**, Presented three lectures on introductory and advanced Numpy

Spring 2019 **Teaching Assistant**, *PHYS 0110: Introduction to Physics 1*, with Prof. Matteo Broccio and Brian Pardo at Univ. of Pittsburgh

Fall 2018 **Teaching Assistant**, *ASTRON0088: From Stonehenge to Hubble*, with Prof. Carles Badenes and Prof. Sandhya Rao at Univ. of Pittsburgh

Software

(List of software packages I am the primary developer of)

- `LAZY`, *Python package to use tabular foundation models using a common API*
- `Cal-PIT`, *Python package to diagnose and recallibrate PDFs to ensure conditional coverage*
- `desigal`, *Python package providing utilities to use DESI spectra for studies of galaxies*
- `spline_basis`, *Python package `B-spline` and `I-spline` basis functions to represent PDFs*

Science Communication

9. *Here be Dragons: Charting the Cosmos.* Astronomy on Tap, Toronto, June, 2026.
8. *A Career in Physics and Interdisciplinary Research.* The Vedam School Yehlanka, Bangalore, India, October, 2025.
7. *Cosmic Cartography: Making the Largest Maps of our Universe.* Allegheny Observatory Public Lecture, Pittsburgh, June, 2024.
6. *Your Guide to the Zodiacs.* Astronomy on Tap, Space Bar, Pittsburgh, February, 2024.
5. *Making the Largest Maps of Our Universe.* Produced an exhibit for the 2022 ACCelerate Creativity + Innovation Festival at the Smithsonian National Museum of American History, April, 2022. Secured funding of ∼$10,000. Event attended by more than 10,000 visitors over 3 days.
4. *How Stars Helped to Build Human Civilizations.* Biophilia Pittsburgh at the Phipps Conservatory and Botanical Gardens, Pittsburgh, November, 2020.
3. *Demystifying Research Internships Abroad: Mitacs Globalink Research Fellowship.* Student Development Council Talk Series, IISER Bhopal, September, 2020.
2. *Mapping the Universe using Sky Surveys.* NISER Astronomy Club Alumni Talk, National Institute of Science Education and Research, August, 2020.
1. *Tutor for DESI High: Enabling high school students to use data from DESI* at the 2020 Bay Area Science Festival, 2021 North Carolina Science Festival, 2021 Boston Science Festival, and DESI High@Nepal 2021.

List of Publications

ADS profile with an up-to-date citation record can be found here.
(12 lead/significant contributing author, and 127 contributing author)

Lead/Significant Contributing Author

12. **Dey, B.**, and Speagle, J. S., 2026, 9th Workshop on Machine Learning and the Physical Sciences (ML4PS 2026; Oral Spotlight),
Tabular Foundation Models Dramatically Outperform Bespoke Photometric Redshift Estimators.
11. Peng, Z.†, **Dey, B.***, Maddison C. J., and Speagle, J. S., 2026, Conference on Physics and AI at Stanford University (PAI 2026),
Integral Field Unit Spectroscopy with One Fiber.
10. Pratsos, A.†, **Dey, B.***, and Li, T. S., 2026, Conference on Physics and AI at Stanford University (PAI 2026),
Characterizing Stellar Streams with Error-Aware Machine Learning.
9. **Dey, B.**, Newman, J. A., Zhang, T.Q., et al., 2026, The Astronomical Journal, 172, 65,
Deep Spectroscopy with DESI for Photometric Redshift Training and Calibration.
8. **Dey, B.**, Zhao, B., Newman, J. A., et al., 2025, Machine Learning for Science and Technology, 6, 045058,

Towards Instance-Wise Calibration: Local Amortized Diagnostics and Reshaping of Conditional Densities (LADaR).

7. Setton, D. J., **Dey, B.**, Khullar, G., et al., 2023, The Astrophysical Journal, 947, L31,
DESI Survey Validation Spectra Reveal an Increasing Fraction of Recently Quenched Galaxies at $z \sim 1$.

6. Zhou, R., **Dey, B.**, Newman, J. A., et al., 2023, The Astronomical Journal, 165, 58,
Target Selection and Validation of DESI Luminous Red Galaxies.

5. **Dey, B.**, Newman J. A., Andrews B. H., et al., 2022, Monthly Notices of the Royal Astronomical Society, 515, 5285,
Photometric redshifts from SDSS images with an interpretable deep capsule network.

4. Chen, T. Y., **Dey, B.***, Ghosh A., et al., 2022, Proceedings of the US Community Study on the Future of Particle Physics (Snowmass 2021).
Interpretable Uncertainty Quantification in AI for HEP.

3. **Dey, B.**, Newman, J.A., Andrews, B.H., et al., 2021, Fourth Workshop on Machine Learning and the Physical Sciences (NeurIPS 2021),
Re-calibrating Photometric Redshift Probability Distributions Using Feature-space Regression.

2. **Dey, B.**, Rosolowsky, E., Cao, Y., et al., 2019, Monthly Notices of the Royal Astronomical Society, 488, 1926,
The EDGE-CALIFA survey: Exploring the star formation law through variable selection.

1. Bhattacharjee, S., **Dey, B.**, Mohapatra, A. K., 2018, European Journal of Physics, 39, 035404,
Study of geometric phase using classical coupled oscillators.

† Student mentee * Corresponding author

Contributing Author

127. Perez-Sanchez, G., et al.[including **Dey, B.**], 2026, Journal of Cosmology and Astroparticle Physics, 2026, 013,
Alcock-Paczynski blinding scheme for the Lyα forest analysis.

126. Chan, J.-K., et al.[including **Dey, B.**], 2026, The Astrophysical Journal, 1011, 110,
Unveiling Hidden Lyman Alpha Emitters in the DESI DR1 Data.

125. Napolitano, L., et al.[including **Dey, B.**], 2026, The Astronomical Journal, 172, 250,
The Composite Spectrum of QSO Absorption Line Systems in DESI DR2.

124. Uberoi, N., et al.[including **Dey, B.**], 2026, Journal of Cosmology and Astroparticle Physics, 2026, 068,
The linear point standard ruler with DESI DR1 and DR2 data.

123. Lange, J. U., et al.[including **Dey, B.**], 2026, The Astrophysical Journal, 1010, 170,
Cosmological Constraints from Full-scale Clustering and Galaxy-Galaxy Lensing with DESI DR1.

122. Ashmead, F., et al.[including **Dey, B.**], 2026, The Astrophysical Journal, 1010, 140,
Optimizing Photometric Redshift Training Sets. I. Efficient Compression of the Galaxy Color-Redshift Relation with UMAP.

121. Carlberg, R. G., et al.[including **Dey, B.**], 2026, The Astrophysical Journal, 1010, 115,
Joint Modeling of GD-1 and C-19 as Old Streams.

120. Scholte, D., et al.[including **Dey, B.**], 2026, Monthly Notices of the Royal Astronomical Society, 551, stag1381,
Electron temperature relations and the direct N, O, Ne, S, and Ar abundances of 49 959 star-forming galaxies in DESI data release 2.

119. Alcolea, J. A., et al.[including **Dey, B.**], 2026, Astronomy and Astrophysics, 713, A133,
Beyond traditional emission-line diagnostics: Using autoencoders to uncover active galactic nuclei in DESI spectra.

118. Morawetz, J., et al.[including **Dey, B.**], 2026, Journal of Cosmology and Astroparticle Physics, 2026, 001,
Frequentist cosmological constraints from full-shape clustering measurements in DESI DR1.

117. Dadiani, E., et al.[including **Dey, B.**], 2026, arXiv e-prints, arXiv:2607.27390,
Cosmic Pairs: A DESI Census of Dual and Offset AGN as Precursors to Massive Black Hole Binaries.

116. Perez-Sanchez, G., et al.[including **Dey, B.**], 2026, arXiv e-prints, arXiv:2607.07875,
Alcock-Paczynski Blinding Scheme for the Ly- Forest Analysis.

115. Rodríguez Morales, V., et al.[including **Dey, B.**], 2026, arXiv e-prints, arXiv:2607.02655,
The largest sample of AGN outflows in dwarf galaxies using DESI DR1.

114. Bonici, M., et al.[including **Dey, B.**], 2026, arXiv e-prints, arXiv:2607.02498,
Alleviating prior dependencies for DESI DR1 clustering fits through reparameterization.

113. Salcedo Hernandez, Y., et al.[including **Dey, B.**], 2026, The Astronomical Journal, 172, 11,
Highly Efficient Selection of High-redshift Emission-line Galaxies for Future DESI-like Surveys with Deep Multiband Imaging.

112. Brown, Z., et al.[including **Dey, B.**], 2026, arXiv e-prints, arXiv:2606.24651,
Measuring local primordial non-Gaussianity from the clustering of DESI DR1 LRGs and QSOs.

111. Forero-Sánchez, D., et al.[including **Dey, B.**], 2026, arXiv e-prints, arXiv:2606.23936,
Cosmological constraints from the DESI DR1 Bispectrum Full-Shape and DR2 BAO.

110. Torres-Gomez, V., et al.[including **Dey, B.**], 2026, arXiv e-prints, arXiv:2606.21035,
Quality Assessment of Spectroscopic Data Reduction Pipelines Using Artificial Intelligence: Scrutinizing Data Release 2 from the DESI Survey.

109. Hagen, T., et al.[including **Dey, B.**], 2026, arXiv e-prints, arXiv:2606.18581,
DESI Data Release 2 ELGs: Property-dependent subsamples, imaging systematics, and clustering.

108. Moore, S. G., et al.[including **Dey, B.**], 2026, Monthly Notices of the Royal Astronomical Society, 549, stag866,
DESI DR2 Galaxy luminosity functions.

107. Dey, A., et al.[including **Dey, B.**], 2026, The Astronomical Journal, 171, 348,
The Backup Program of the Dark Energy Spectroscopic Instrument's Milky Way Survey.

106. Khederlarian, A., et al.[including **Dey, B.**], 2026, The Astronomical Journal, 171, 340,
Optimizing Deep Learning Photometric Redshifts for the Roman Space Telescope with HST/CANDELS.

105. Semenaite, A., et al.[including **Dey, B.**], 2026, The Open Journal of Astrophysics, 9, 62334,
Joint cosmological fits to DESI-DR1 full-shape clustering and weak gravitational lensing in configuration space.

104. Nguyen, A., et al.[including **Dey, B.**], 2026, The Open Journal of Astrophysics, 9, 61766,
Detection of supernova magnitude fluctuations induced by large-scale structure.

103. Turner, W., et al.[including **Dey, B.**], 2026, Journal of Cosmology and Astroparticle Physics, 2026, 010,
Probing the limits of cosmological information from the Lyman-α forest 2-point correlation functions.

102. Chen, X., et al.[including **Dey, B.**], 2026, Journal of Cosmology and Astroparticle Physics, 2026, 001,
Extensive analysis of reconstruction algorithms for DESI 2024 baryon acoustic oscillations.

101. DESI Collaboration, et al.[including **Dey, B.**], 2026, The Astronomical Journal, 171, 285,
Data Release 1 of the Dark Energy Spectroscopic Instrument.

100. Jarvis, E., et al.[including **Dey, B.**], 2026, arXiv e-prints, arXiv:2604.20958,
Characterizing the GD-1 Stream with DESI DR2 Data: Thin Stream and Hot Cocoon.

99. Bansal, P., et al.[including **Dey, B.**], 2026, arXiv e-prints, arXiv:2604.08895,
FolpsD: combining EFT and phenomenological approaches for joint power spectrum and bispectrum analyses.

98. Rosado-Marín, A. J., et al.[including **Dey, B.**], 2026, arXiv e-prints, arXiv:2604.05213,
Local primordial non-Gaussianity using cross-correlations of DESI tracers.

97. Chaussidon, E., et al.[including **Dey, B.**], 2026, arXiv e-prints, arXiv:2604.04867,
Measurement of the galaxy-velocity power spectrum of DESI tracers with the kinematic Sunyaev-Zeldovich effect using DESI DR2 and ACT DR6.

96. Porredon, A. ,., et al.[including **Dey, B.**], 2026, The Open Journal of Astrophysics, 9, 61342,
DESI-DR1 3 × 2 -pt analysis: consistent cosmology across weak lensing surveys.

95. Valcin, D., et al.[including **Dey, B.**], 2026, Journal of Cosmology and Astroparticle Physics, 2026, 058,
Combined tracer analysis for DESI 2024 BAO.

94. Karaçaylı, N. G., et al.[including **Dey, B.**], 2026, Journal of Cosmology and Astroparticle Physics, 2026, 048,
DESI DR1 Lyα 1D power spectrum: Validation of estimators.

93. Amsellem, A. J., et al.[including **Dey, B.**], 2026, The Astrophysical Journal, 1001, 157,
Probing the Environment around GW170817 with DESI: Insights on Galaxy Group Peculiar Velocities for Standard Siren Measurements.

92. Wang, Y., et al.[including **Dey, B.**], 2026, arXiv e-prints, arXiv:2603.25693,
Nonlinear Information from DESI Luminous Red Galaxies: An Emulator-Based Analysis of Pre- and Post-Reconstruction Power Spectra.

91. Emas, N. P. A. P., et al.[including **Dey, B.**], 2026, The Open Journal of Astrophysics, 9, 59886,
Validation of the DESI-DR1 3x2-pt analysis: scale cut and shear ratio tests.

90. Ebina, H., et al.[including **Dey, B.**], 2026, Journal of Cosmology and Astroparticle Physics, 2026, 019,
Clustering analysis of medium-band selected high-redshift galaxies.

89. Wang, H., et al.[including **Dey, B.**], 2026, Monthly Notices of the Royal Astronomical Society, 545, staf2069,
Galaxy-multiplet clustering from DESI DR2.

88. Adame, A. G., et al.[including **Dey, B.**], 2026, Journal of Cosmology and Astroparticle Physics, 2026, E02,
Erratum: DESI 2024 V: Full-Shape Galaxy Clustering from Galaxies and Quasars.

87. Hsu, Y.-M., et al.[including **Dey, B.**], 2026, The Astrophysical Journal Supplement Series, 282, 41,
A New Way to Discover Strong Gravitational Lenses: Pairwise Spectroscopic Search from DESI DR1.

86. Ratajczak, J., et al.[including **Dey, B.**], 2026, The Astronomical Journal, 171, 71,
The Compilation and Validation of the Spectroscopic Redshift Catalogs for the DESI-COSMOS and DESI-XMM-LSS Fields.

85. Casas, L., et al.[including **Dey, B.**], 2026, Physical Review D, 113, 023520,
Validation of the DESI DR2 Lyα BAO analysis using synthetic datasets.

84. Koposov, S., et al.[including **Dey, B.**], 2026, The Open Journal of Astrophysics, 9, 55260,
DESI Data Release 1: Stellar Catalogue.

83. Gordon, C., et al.[including **Dey, B.**], 2026, Monthly Notices of the Royal Astronomical Society, 545, staf2035,
Modelling the impact of quasar redshift errors on the full-shape analysis of correlations in the Lyman-α forest.

82. Moran, E. R., et al.[including **Dey, B.**], 2026, The Astrophysical Journal, 997, 36,
Deep Learning Improves Photometric Redshifts in All Regions of Color Space.

81. Ruggeri, R., et al.[including **Dey, B.**], 2025, arXiv e-prints, arXiv:2512.15963,
Clustering redshift distribution calibration of weak lensing surveys using the DESI-DR1 spectroscopic dataset.

80. Nagaraj, G., et al.[including **Dey, B.**], 2025, The Astrophysical Journal, 995, 126,
ODIN: Probing the LAE Lyα Luminosity Function across Cosmic Time and Different Environments.

79. Kumwembe, A. A., et al.[including **Dey, B.**], 2025, Research Notes of the American Astronomical Society, 9, 305,
Enhancing Multiplet Alignment Measurements with Imaging.

78. Brown, Z., et al.[including **Dey, B.**], 2025, Monthly Notices of the Royal Astronomical Society, 543, 2078,
Constraining primordial non-Gaussianity from the large scale structure two-point and three-point correlation functions.

77. Garcia-Quintero, C., et al.[including **Dey, B.**], 2025, Physical Review D, 112, 083529,
Cosmological implications of DESI DR2 BAO measurements in light of the latest ACT DR6 CMB data.

76. Abdul Karim, M., et al.[including **Dey, B.**], 2025, Physical Review D, 112, 083515,
DESI DR2 results. II. Measurements of baryon acoustic oscillations and cosmological constraints.

75. Abdul Karim, M., et al.[including **Dey, B.**], 2025, Physical Review D, 112, 083514,
DESI DR2 results. I. Baryon acoustic oscillations from the Lyman alpha forest.

74. Elbers, W., et al.[including **Dey, B.**], 2025, Physical Review D, 112, 083513,
Constraints on neutrino physics from DESI DR2 BAO and DR1 full shape.

73. Andrade, U., et al.[including **Dey, B.**], 2025, Physical Review D, 112, 083512,
Validation of the DESI DR2 measurements of baryon acoustic oscillations from galaxies and quasars.

72. Lodha, K., et al.[including **Dey, B.**], 2025, Physical Review D, 112, 083511,
Extended dark energy analysis using DESI DR2 BAO measurements.

71. Brodzeller, A., et al.[including **Dey, B.**], 2025, Physical Review D, 112, 083510,
Construction of the damped Lyα absorber catalog for DESI DR2 Lyα BAO.

70. Hadzhiyska, B., et al.[including **Dey, B.**], 2025, Physical Review D, 112, 083509,
Evidence for large baryonic feedback at low and intermediate redshifts from kinematic Sunyaev-Zel'dovich observations with ACT and DESI photometric galaxies.

69. de Belsunce, R., et al.[including **Dey, B.**], 2025, Journal of Cosmology and Astroparticle Physics, 2025, 077,
Cosmology from Planck CMB lensing and DESI DR1 quasar tomography.

68. Karaçaylı, N. G., et al.[including **Dey, B.**], 2025, Journal of Cosmology and Astroparticle Physics, 2025, 004,
DESI DR1 Lyα 1D power spectrum: the optimal estimator measurement.

67. Treiber, H., et al.[including **Dey, B.**], 2025, arXiv e-prints, arXiv:2509.20434,
Dwarf galaxy halo masses from spectroscopic and photometric lensing in DESI and DES.

66. Adame, A. G., et al.[including **Dey, B.**], 2025, Journal of Cosmology and Astroparticle Physics, 2025, 008,
DESI 2024 V: Full-Shape galaxy clustering from galaxies and quasars.

65. Pucha, R., et al.[including **Dey, B.**], 2025, The Astrophysical Journal, 990, 231,
Erratum: "Tripling the Census of Dwarf AGN Candidates Using DESI Early Data" (2025, ApJ, 982, 10).

64. Slepian, Z., et al.[including **Dey, B.**], 2025, arXiv e-prints, arXiv:2508.09133,
Measurement of Parity-Violating Modes of the Dark Energy Spectroscopic Instrument (DESI) Year 1 Luminous Red Galaxies' 4-Point Correlation Function.

63. Ahlen, S. P., et al.[including **Dey, B.**], 2025, Physical Review Letters, 135, 081003,
Positive Neutrino Masses with DESI DR2 via Matter Conversion to Dark Energy.

62. Zhou, R., et al.[including **Dey, B.**], 2025, The Open Journal of Astrophysics, 8, 83,
Stellar reddening map from DESI imaging and spectroscopy.

61. Adame, A. G., et al.[including **Dey, B.**], 2025, Journal of Cosmology and Astroparticle Physics, 2025, 028,
DESI 2024 VII: cosmological constraints from the full-shape modeling of clustering measurements.

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The clustering of Lyman Alpha Emitting galaxies at =2 3.

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The Early Data Release of the Dark Energy Spectroscopic Instrument.

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The CluMPR galaxy cluster-finding algorithm and DESI legacy survey galaxy cluster catalogue.

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Emission line predictions for mock galaxy catalogues: a new differentiable and empirical mapping from DESI.

20. Lamman, C., et al.[including **Dey, B.**], 2024, Monthly Notices of the Royal Astronomical Society, 528, 6559,
Redshift-dependent RSD bias from intrinsic alignment with DESI Year 1 spectra.

19. DESI Collaboration, et al.[including **Dey, B.**], 2024, The Astronomical Journal, 167, 62,
Validation of the Scientific Program for the Dark Energy Spectroscopic Instrument.

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17. Moustakas, J., et al.[including **Dey, B.**], 2023, Astrophysics Source Code Library, ascl:2308.005,
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NANCY: Next-generation All-sky Near-infrared Community surveY.

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The DESI Bright Galaxy Survey: Final Target Selection, Design, and Validation.

14. Guy, J., et al.[including **Dey, B.**], 2023, The Astronomical Journal, 165, 144,
The Spectroscopic Data Processing Pipeline for the Dark Energy Spectroscopic Instrument.

13. Raichoor, A., et al.[including **Dey, B.**], 2023, The Astronomical Journal, 165, 126,
Target Selection and Validation of DESI Emission Line Galaxies.

12. Alexander, D. M., et al.[including **Dey, B.**], 2023, The Astronomical Journal, 165, 124,
The DESI Survey Validation: Results from Visual Inspection of the Quasar Survey Spectra.

11. Chaussidon, E., et al.[including **Dey, B.**], 2023, The Astrophysical Journal, 944, 107,
Target Selection and Validation of DESI Quasars.

10. Myers, A. D., et al.[including **Dey, B.**], 2023, The Astronomical Journal, 165, 50,
The Target-selection Pipeline for the Dark Energy Spectroscopic Instrument.

9. Lan, T.-W., et al.[including **Dey, B.**], 2023, The Astrophysical Journal, 943, 68,
The DESI Survey Validation: Results from Visual Inspection of Bright Galaxies, Luminous Red Galaxies, and Emission-line Galaxies.

8. Silber, J. H., et al.[including **Dey, B.**], 2023, The Astronomical Journal, 165, 9,
The Robotic Multiobject Focal Plane System of the Dark Energy Spectroscopic Instrument (DESI).

7. DESI Collaboration, et al.[including **Dey, B.**], 2022, The Astronomical Journal, 164, 207,
Overview of the Instrumentation for the Dark Energy Spectroscopic Instrument.

6. Setton, D., et al.[including **Dey, B.**], 2022, HST Proposal, 17110,
Post-starbursts from DESI: Timing quenching and morphological transformation at $1 < z < 1.3$.

5. Sand, K. R., et al.[including **Dey, B.**], 2022, The Astrophysical Journal, 932, 98,
Multiband Detection of Repeating FRB 20180916B.

4. Barth, M., et al.[including **Dey, B.**], 2022, American Astronomical Society Meeting #240, 240, 139.10,
A new galaxy cluster-finding algorithm.

3. Zhou, R., et al.[including **Dey, B.**], 2020, Research Notes of the American Astronomical Society, 4, 181,
Preliminary Target Selection for the DESI Luminous Red Galaxy (LRG) Sample.

2. Sand, K. R., et al.[including **Dey, B.**], 2020, The Astronomer's Telegram, 13781, 1,
Low-frequency detection of FRB180916 with the uGMRT.

1. Dutta, S., Khandai, N., Dey, B., 2020, Monthly Notices of the Royal Astronomical Society, 494, 2664,
The population of galaxies that contribute to the H I mass function.

List of Presentations

(27 Invited and 21 Contributed Presentations)

Invited

27. *Photometric Redshifts for Next-Generation Sky Surveys.* CASSA Colloquium, Independent University of Bangladesh, Bangladesh, November, 2025.

26. *Photometric Redshifts for Next-Generation Sky Surveys.* CITA Seminar, Canadian Institute for Theoretical Astrophysics, University of Toronto, Canada, September 2025.

25. *Photometric Redshifts for Next-Generation Sky Surveys.* TASTY talks, Department of Astronomy and Astrophysics, University of Toronto, Canada, November 2024.

24. *Photometric Redshifts for Next-Generation Sky Surveys.* Astrophysics Seminar, Université de Montréal, Canada, October 2024.

23. *Photometric Redshifts for Next-Generation Sky Surveys.* Cosmology Group Meeting, Perimeter Institute, Canada, October 2024.

22. *DESI Spectroscopy for Photo-z Training and Calibration.* Plenary Talk at Winter 2023 DESI Collaboration Meeting, Hawaii, USA, December 2023.

21. *Photometric Redshifts for Next-Generation Sky Surveys.* Cosmology X Data Science Meeting, Center for Computational Astrophysics, Flatiron Institute, USA, November 2023.

20. *Photometric Redshifts for Next-Generation Sky Surveys.* Yale Cosmology Seminar, Yale University, USA, November 2023.

19. *Photometric Redshifts for Next-Generation Sky Surveys.* Survey Science Meeting, Princeton University, USA, November 2023.

18. *Photometric Redshifts for Next-Generation Sky Surveys.* Astrolunch seminar, University of Pittsburgh, USA, October 2023.

17. *Photometric Redshifts for Next-Generation Sky Surveys.* CCAPP Seminar, The Ohio State University, USA, October 2023.

16. *Photometric Redshifts for Next-Generation Sky Surveys.* JPL Dark Sector Meeting, NASA Jet propulsion Laboratory, USA, September 2023.

15. *Photometric Redshifts for Next-Generation Sky Surveys.* Caltech/IPAC Lunch Seminar, Infrared Processing & Analysis Center (IPAC), Pasadena, USA, September 2023.

14. *The DESI Photometric Redshift Topical Group*, Plenary talk at the 2023 Summer DESI Collaboration Meeting. Durham University, Durham, UK July 2023.

13. *Photometric Redshifts using Interpretable Deep Capsule Networks.* Talk at the DESI@UCL symposium, University College London, London, UK, July 2023.

12. *Photometric Redshifts using Interpretable Deep Capsule Networks.* Tea Talk, Kavli Institute for Particle Astrophysics and Cosmology, Stanford University, USA, April 2023.

11. *Calibrated Predictive Distributions for Photometric Redshifts.* Building a physical understanding of galaxy evolution with data-driven astronomy, Kavli Institute for Theoretical Physics, USA, February 2023.

10. *Calibrated Predictive Distributions.* NSF AI Planning Institute for Data-Driven Discovery in Physics, Carnegie Mellon University, USA, September 2022.

9. *Photometric redshifts for next generation sky surveys.* STAtistical Methods for the Physical Sciences (STAMPS) meeting, Carnegie Mellon University, USA, February 2022.

8. *The Dark Energy Spectroscopic Instrument: One year and 13 million redshifts later.* Plenary talk at Summer 2022 LSST-DESC Collaboration meeting at Kavli Institute for Cosmological Physics, University of Chicago, Chicago, USA, August 2022.

7. *Photometric redshifts for next-generation sky surveys.* Talk at Astro-Data group meeting, Princeton University, USA, July 2022.

6. *Photometric redshifts for next-generation sky surveys.* FLASH Lunch talk, University of California, Santa Cruz, USA, June 2022.

5. *Beyond DESI: Making an even larger map of the Universe.* DESI Lunch, Lawrence Berkeley National Laboratory, USA, June 2022.

4. *Photometric Redshifts for Next Generation Sky Surveys.* STAtistical Methods for the Physical Sciences (STAMPS) meeting, Carnegie Mellon University, USA, February 2022.

3. *Photometric Redshifts using Interpretable Deep Capsule Networks.* Institute seminar, Inter-University Centre for Astronomy and Astrophysics (IUCAA), India, December 2021.

2. *Capsule Networks: An Astronomer's Perspective.* Break-out session on Deep Learning, Statistical Challenges in Modern Astronomy (SCMA) VII, June 2021.

1. *Reducing Photometric Redshift Outliers with Deep Learning.* STAtistical Methods for the Physical Sciences (STAMPS) meeting, Carnegie Mellon University, USA, April 2020.

Contributed

21. *Uncertainty Quantification for the Physical Sciences.* Brown Bag Seminar, Dept. of Statistical Sciences, University of Toronto, Canada, September 2024.

20. *Calibrated predictive distributions for photometric redshifts*, Talk at STATSTRO 2024: The AIstronomy Revolution, Toronto, Canada, May 2024.

19. *Calibrated predictive distributions for photometric redshifts*, Talk at the Summer 2024 LSST-ISSC Collaboration Meeting, Boston, USA, April 2024.

18. *Photometric Redshifts for Next Generation of Sky Surveys*, Thesis Talk at the 243rd Meeting of the American Astronomical Society, New Orleans, USA, January 2023.

17. *DESI for photo-z Training and Calibration*, Talk at the 2023 Summer DESC Collaboration Meeting. SLAC National Accelerator Laboratory, USA, July 2023.

16. *Calibrated predictive distributions for photometric redshifts*, Talk at Statistical Challenges in Modern Astronomy (SCMA) VIII, Pennsylvania State University, State College, USA, June 2023.

15. *DESI Deep Spectroscopy for Photo-z Training and Calibration.* Talk at DESI-2/ Stage-5 Workshop, Napa, USA, March 2023.

14. *Stellar Masses using Random Forests.* Talk at DESI Collaboration Meeting, Cancun, Mexico, December 2022.

13. *Calibration of Individual Photometric Redshift Estimates.* Talk at Essential Cosmology for the Next Generation VIII (Cosmology on the Beach), Playa De Carmen, Mexico, November 2022.

12. *The Dark Energy Spectroscopic Instrument: One year and 13 million redshifts later*. Astrosnacks presentation, University of Pittsburgh, September 2022.

11. *Calibrated Probability Distributions for Photometric Redshifts.* Poster at Rubin Observatory Project and Community Workshop, Tucson, USA, August 2022.

10. *Calibrated Probability Distributions for Photometric Redshifts.* Poster and Talk at Summer 2022 LSST-DESC Collaboration meeting at Kavli Institute for Cosmological Physics, University of Chicago, Chicago, USA, August 2022.

9. *Calibrated Predictive Distributions for Photometric Redshifts.* Poster at ICML 2022 Workshop on Machine Learning for Astrophysics, Baltimore, USA, July, 2022.

8. *Recalibrating Probability Density Estimates Using Feature-Space Regression.* Refereed talk at the Symposium on Data Science and Statistics, Pittsburgh, USA, June 2022.

7. *Re-calibrating Photometric Redshift Probability Distributions Using Feature-space Regression.* Poster and Talk at the Fourth Workshop on Machine Learning and the Physical Sciences (NeurIPS 2021), December 2021.

6. *Interpretable Photometric Redshifts using Deep Capsule Networks.* Talk at the 2nd Symposium on Artificial Intelligence for Science, Industry, and Society (AISIS 2021), October 2021.

5. *Latent Variable Models: Principal Components.* Talk at AstroPGH-TAMU Bootcamp 2021.

4. *Interpretable Photometric Redshifts with a DeepCapsule Network.* Poster at Statistical Challenges in Modern Astronomy VII, June 2021.

3. *Mapping the Universe using Sky Surveys.* Astrosnacks presentation, University of Pittsburgh, July 2020.

2. *Ancillary Targets: Testing filler samples in Survey validation.* DESI Collaboration meeting, Ohio State University, December 2019.

1. *LRG & ELG Imaging systematic Trends* (with A. Raichoor). DESI virtual collaboration meeting, March 2020.

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